

1.BRAINSTORMING AND IDEATION PHASE

The Brainstorming and Ideation Phase is the initial stage of the **FitBuddy – AI Fitness Plan Generator Using Gemini** project. The main purpose of this phase is to identify a practical problem in the fitness and wellness domain and develop an innovative technology-based solution using Generative Artificial Intelligence.

Many people want to improve their fitness but may not know how to create a suitable workout routine. Traditional fitness applications often provide predefined workout plans that may not sufficiently consider an individual's age, weight, fitness goal, preferred workout intensity, and feedback. Therefore, the idea of FitBuddy was developed to provide a more personalized and interactive fitness planning experience.

FitBuddy is designed as a web-based AI application that uses Google's Gemini Generative AI technology to generate personalized seven-day workout plans. The system collects basic information from the user, such as name, age, weight, fitness goal, and workout intensity. This information is processed and provided to the AI model, which generates an appropriate workout plan.

The brainstorming process also identified the importance of allowing users to provide feedback after receiving a generated plan. Instead of generating a fixed plan, FitBuddy provides a feedback mechanism through which users can request changes. The AI system can then generate an updated workout plan according to the feedback.

Another important idea identified during brainstorming was the inclusion of nutrition and recovery guidance. The project therefore includes an AI-generated nutrition or recovery tip along with the workout plan. The application is intended to provide general wellness information rather than medical diagnosis or treatment.

The major ideas identified during this phase were:

- AI-based personalized workout generation.
- Seven-day workout planning.
- User-specific fitness goals.
- Adjustable workout intensity.
- AI-generated nutrition and recovery tips.
- Feedback-based workout plan modification.
- Database storage of user information and generated plans.
- Administrator access for viewing stored users and plans.
- Simple and responsive web interface.

- REST API support for future expansion.

The brainstorming phase therefore established the foundation of FitBuddy and provided the direction for the remaining development phases.